

Research article

Socio-demographic disparities in climate change anxiety among British Columbians impacted by successive extreme weather events

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ABSTRACT

Introduction: This study aimed to identify socio-demographic factors associated with climate change anxiety (CCA) in British Columbia, Canada, and examined the effect of compounding climate-related events in 2021 on CCA levels among historically marginalized populations.

Methods: Participants were recruited via social media across three cross-sectional survey waves from May to December 2021. A multivariate logistic regression model explored the association between moderate/high vs low/no CCA and socio-demographic characteristics, including age, gender, ethnicity, disability, sexual orientation, relationship status, occupation, income, education, geography, and political orientation. We tested for interaction effects between the survey wave and each socio-demographic factor to identify sub-populations with increasing CCA scores over time.

Results: Among 1179 participants, the median CCA score was 1.46 (range 1–5). The multivariable model revealed greater odds of moderate/high CCA among women, individuals unsure whether they wanted children, those with a Bachelor's degree or higher, and those with a liberal political orientation. Individuals who were older, higher-income, and not in a relationship had lower odds of moderate/high CCA. Interaction effects revealed significant differences in CCA trends over time for most groups, except for ethnicity and relationship status. CCA increased over time among non-binary individuals, Indigenous participants, those who were slightly conservative, and those educated at a high school level or less.

Conclusions: This study revealed specific facets of social categories associated with increased CCA and identified disparities in CCA distribution and temporal patterns following exposure to successive extreme weather events. To effectively address CCA, tailored mental health interventions must be developed across diverse social categories.

1. Introduction

Climate change, a global health emergency [1,2], has implications for human well-being and ecological systems, particularly due to the rising frequency, severity, and duration of extreme weather events such as heatwaves, droughts, and floods [3]. Successively occurring extreme weather events are influenced by societal or environmental risks [4]. Canada has also emitted disproportionate levels of emissions for its

relatively small population, and as a relatively rich country, it is within its abilities to lower its emissions and support developing countries that cannot [5].

Evidence suggests that climate change and successive extreme weather events can have multifaceted mental health impacts, exacerbating pre-existing conditions or triggering acute and chronic psychological consequences, including climate change anxiety (CCA) [6,7]. CCA, previously described as the persistent fear of environmental doom,

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is a concern that describes the emotional distress resulting from witnessing or learning about the devastating impacts of climate change on environmental and human health [6]. The impacts of climate change associated with CCA can be felt through environmental stressors, particularly direct experiences (leading to immediate shocks or post-traumatic stress disorder; PTSD and secondary pathways (through disruptions to critical infrastructure, leading to loss of essentials (e.g. livelihoods, homes), resulting in mental health conditions including depression, anxiety, and suicide ideation) [6–8].

Despite the recognized link between climate change and mental health, research into the factors predisposing populations to heightened vulnerability remains scarce. The effects of climate change are not equitably distributed, with historically marginalized populations (e.g., older adults [9], low socioeconomic status (SES) individuals [6], and Indigenous populations [10]) often bearing disproportionate levels of direct and indirect climate change harms, and, consequently, mental health impacts [6,10–12]. Pathways for these inequities include socio-cultural, economic, educational and environmental circumstances [7].

Notably, identifying socio-demographic nuances shaping CCA can offer insights into the intersecting vulnerabilities faced by different populations. Intersectionality, or the ongoing tension between social identities, social locations, and structural factors, is key when tailoring interventions and initiatives [13]. Such insights can inform mental health interventions to safeguard mental health, tailored climate policies, and community engagement strategies in highly susceptible climate-affected contexts.

Recent research found an increase in CCA levels after the 2021 heat wave in British Columbia, Canada [14]. Key gaps remain, however, regarding a) factors that influence mental health vulnerability to successive extreme weather events, b) potential compounding mental health risks over extreme weather events and/or time, and c) socio-demographic differences in CCA trajectories over time [14].

Addressing these gaps, this study aimed to investigate socio-demographic factors associated with CCA trends, in British Columbia (BC) following the Western North America heat wave (hereafter “the heat dome”), and the November atmospheric river flooding in BC (hereafter “the floods”). These two consecutive extreme weather events occurred in BC in mid- to late 2021. BC, a region in Canada, often experiences extreme events, but is seeing alarming trends including hotter summers, increased precipitation, more frequent forest fires, and floods [15]. The heat dome encompassed most of the Pacific Northwest of North America between June 25th and July 1st, with temperatures reaching 49.6°C (121.3 °F), was associated with wildfires that destroyed entire communities, including Lytton, BC, and was linked to over 750 excess deaths [14,16,17]. The floods occurred due to two atmospheric rivers between November 11th and 16th, 2021 [18]. Over 1000 homes were flooded in Abbotsford, BC, five people were killed due to a resulting landslide, and infrastructure and agriculture was destroyed in Southern BC, with insurable and non-insurable damage costing \$675 million and \$4.7 billion, respectively [16,19,20]. Additionally, in 2023, extreme wildfires occurred throughout BC, with total costs exceeding \$1 billion, and six wildland firefighters lost their lives [21]. Wildfires and the resulting evacuations disproportionately impact First Nations communities; between 2009 and 2019, BC had 253 fires requiring evacuation, and despite being only 6% of the population, 26% of wildfire evacuees were Indigenous [22].

2. Materials and methods

2.1. Study design and data collection

The study employed a repeated cross-sectional, quasi-experimental design with three survey waves, and analyzed pooled data from the BC Climate Distress Monitoring System (CDMS) survey. The CDMS, described in more detail previously, recruited British Columbian

residents aged 16 years and older using paid social media advertisements on Facebook and Instagram [14]. The full survey can be accessed at <https://mhcca.ca/datasets>. CDMS data were collected in three survey waves between May and December 2021. The first wave (May 12th - June 21st) preceded the 2021 heat dome, the second wave (July 1st - July 18th) followed the heat dome, and the third wave (November 30th - December 4th) occurred after the 2021 floods. Due to the serial cross-sectional nature of the study, each participant was surveyed in a single wave. Eligible participants provided informed consent and completed a 10-minute online survey using the SurveyMonkey platform (<http://www.surveymonkey.com>). Analytic exclusion criteria further restricted the sample to individuals with complete data across the variables included in the analysis. This study also used other elements of the CDMS, including psychological scales and demographic information.

2.2. Variables of interest

The *outcome variable* in this analysis was moderate to high CCA, as measured by Clayton and Karaszia's Climate Change Anxiety Scale (CCAS) [23]. The full CCAS is a 22-item tool with 4-subscales. This study used the validated 13-item version of the CCAS, which measures self-reported emotional responses to climate change, specifically CCA, on a 5-point Likert scale ranging from 1 (low CCA) to 5 (high CCA). The scale includes a Cognitive Subscale (items 1 - 8; e.g., “I find myself crying because of climate change”) and a Functional Subscale (items 9 - 13; e.g., “My concerns about climate change undermine my ability to work”). Final CCAS scores are calculated as the average score of scale items. Due to the skewed distribution of CCAS scores in our sample and to ensure adequate sensitivity for “high” CCA as a distress measure, a dichotomized CCA variable was created using a cut-off point of 1.85 to indicate moderate to high CCA (“low/none” [CCAS scores <1.85] vs “moderate/high” [CCAS scores ≥ 1.85]). The optimal cut-off point for CCA was determined using the ‘cutpointr’ package in R, which maximizes the scale's sensitivity and specificity in predicting generalized serious psychological distress, as identified by a total score of 13 or higher on the Kessler Psychological Distress Scale (K6) [24–26]. The K6 (also included in the CDMS survey) is a validated 6-item scale assessing the frequency and persistence of non-specific psychological distress symptoms (e.g., “Felt hopeless”) on a 5-point Likert scale (0 = None of the time to 4 = All of the time). Final K6 scores are generated by summing individual item responses, ranging from 0 (low distress) to 24 (high distress). Cut point analyses indicated an optimal CCA cutpoint score of 1.85 or higher (Area under the ROC Curve [AUC] = 0.74, Sensitivity = 0.75, Specificity = 0.73). This is in line with previous research that has sought to identify clinical cutoff points for the CCAS, identifying scores of 21 for mild to moderate climate anxiety and 23 for severe to extremely severe climate anxiety [27]. Our sample has a Cronbach's alpha of 0.907 with the CCAS, showing strong reliability.

Covariates of interest were selected based on a preliminary literature search [28–33] and are shown in Table A.1.

2.3. Data analysis

Statistical analyses were conducted in R (version 4.2.1) [34]. Frequencies (n) and proportions (%) were reported for categorical variables, while means and standard deviations (SD) were used for continuous variables. Bivariate analyses were stratified by CCA severity level (<1.85 or ≥1.85) and survey wave. To assess bivariate differences, we used Chi-Squared tests (χ^2) for categorical variables, one-way analysis of variance (ANOVA) tests for continuous normal variables, and Kruskal-Wallis tests for continuous non-normal variables.

A multivariate logistic regression model assessed associations between socio-demographic factors and moderate/high CCA. Model covariates included survey wave, age, gender, ethnicity, disability status, sexual orientation, relationship status, child-raising intentions, occupation, income, education, population density of residential area,

and political orientation.

To explore trends in CCA among specific sub-populations over time, we examined multiplicative interactions between socio-demographic factors and survey waves on CCA severity levels (<1.85 or ≥ 1.85). We built separate regression models to assess interaction effects, including interaction terms between a continuous numeric measure indicating survey wave and each socio-demographic factor of interest. Additionally, we graphically assessed the distribution and trends in CCAS data across specific sub-populations and survey waves using boxplots with overlaid jitter points to visualize data dispersion. Smoothed loess regression curves were also applied to illustrate underlying trends in CCA for each sub-population across different survey waves. The survey wave was treated as a categorical variable in these boxplots for ease of visual interpretation.

This study utilized pre-existing data from the CDMS dataset. The CDMS power analysis is further described in previous research [14]. A post hoc power analysis is included (Appendix, S1).

2.4. Ethics

Ethical approval for the study, including amendments to use the survey for the second and third waves of data collection, was received from the Research Ethics Board (REB) at Simon Fraser University (REB#: 30000309).

3. Results

Across the three CDMS data collection stages, the study involved 1704 participants. After excluding respondents with incomplete data on the variables of interest ($n = 525$), the analytic sample included 1179 participants with a mean CCAS score of 1.75 (SD = 0.82) and 37.2% exhibiting moderate to high CCA. Participants were approximately evenly distributed across survey waves (wave 1: 36.7%; wave 2: 31.1%; wave 3: 32.1%). Socio-demographic characteristics revealed a nearly equal representation of men (49.3%) and women (46.4%), with most participants identifying as White (82.0%), aged between 25–64 years (70.8%), heterosexual (76.3%), non-disabled (85.6%), in a relationship (73.7%), and earning an annual income between \$0 and \$59,999 (64.6%). Around half of the participants were educated at or above a Bachelor's degree level (48.5%), extremely or moderately liberal (50.6%), and residing in low-density population areas (49.5%).

Bivariate analysis results (Appendix, Table A.2) indicated that participants with moderate/high CCA were most likely to have completed the survey in wave 2, be aged 44 or younger, be non-binary or women, have a disability, be sexually diverse, do not currently have children, work in Arts/Education/Government or Science/Health, earn less than \$60,000, have a Bachelor's degree or higher, live in high-density areas, and have non-specific serious psychological distress.

Across survey waves, bivariate statistics results (Appendix, Table A.3) revealed significant differences based on participants' mean CCAS scores ($p < 0.001$), mean K6 scores ($p < 0.001$), ethnicity ($p = 0.014$), occupation ($p < 0.001$), child-raising intentions ($p = 0.021$), mean political orientation score ($p = 0.004$), and population density ($p = 0.005$).

Multivariable logistic regression results (Table 1) indicated significantly higher odds of experiencing moderate/high CCA (CCAS scores ≥ 1.85) vs low/no CCA among respondents included in survey wave 2 (adjusted OR [aOR]=1.89, [95%CI = 1.32–2.72]), women (aOR=1.42, [95%CI = 1.04–1.94]), individuals unsure about having children (aOR=1.71, [95%CI = 1.02–2.88]), those with at least a Bachelor's degree (aOR=1.80, [95%CI = 1.15–2.85]), and having a more liberal political orientation (per 1-point increase towards extremely liberal on a 7-point political orientation scale; aOR=1.65, [95%CI = 1.49–1.82]). The analysis also found significantly lower odds of moderate/high CCA were associated with older age (65+ years, aOR=0.30, [95%CI = 0.15–0.60]), higher annual income (\$60,000 - \$89,999; aOR=0.51,

Table 1

Multivariable logistic regression estimates of the association between socio-demographic factors and climate change anxiety level of severity among the British Columbians included in the analytic sample ($n = 1179$).

Covariates		aOR	95 %CI	p-value
Survey Wave	Wave 1	reference		
	Wave 2	1.89*	1.32 - 2.72	<0.001
	Wave 3	1.38	0.96 - 1.99	0.085
Age	16–24	reference		
	25–44	1.00	0.61 - 1.63	0.985
	45–64	0.62	0.35 - 1.12	0.115
	65+	0.30*	0.15 - 0.60	<0.001
Gender	Man	reference		
	Non-binary	1.76	0.86 - 3.70	0.129
	Woman	1.42*	1.04 - 1.94	0.026
Ethnicity	White	reference		
	Chinese	0.56	0.24 - 1.28	0.175
	Indigenous	1.39	0.70 - 2.75	0.339
	South Asian	0.60	0.20 - 1.73	0.347
	Other	0.57*	0.33 - 0.97	0.041
Relationship status	Yes	reference		
	No	0.63*	0.45 - 0.88	0.007
Children	Yes	reference		
	No, and I am not sure	1.71	1.02 - 2.88	0.043
	No, and I do not plan to	1.13	0.77 - 1.65	0.547
	No, but I plan to	0.87	0.51 - 1.48	0.600
Income	Less than \$30,000	reference		
	\$30,000 to \$59,999	0.80	0.55 - 1.16	0.231
	\$60,000 to \$89,999	0.51*	0.33 - 0.80	0.003
	\$90,000 or more	0.35*	0.21 - 0.57	<0.001
Education	High School or Less	reference		
	Some Post-Secondary Training	1.28	0.81 - 2.06	0.298
	Bachelor's degree or higher	1.80*	1.15 - 2.85	0.010
Political orientation scores		1.65*	1.49 - 1.82	<0.001

a Wave 1 occurred between May 12th, 2021- June 21st, 2021; wave 2 occurred between July 15th, 2021- July 18th, 2021, and wave 3 occurred between November 6th, 2021- December 1st, 2021.

b Measured per 100 people per square kilometre (km2).

c Measured on a 7-Likert bipolar political orientation scale ranging from 1 ("Extremely Conservative") to 7 ("Extremely Liberal").

* Numbers in bold are statistically significant to a p-value smaller than 0.05.

Note: Socio-demographic covariates included in the model were: survey wave, age, gender, ethnicity, disability status, sexual orientation, relationship status, child-raising intentions, occupation, income, education, population density of residential area, and political orientation.

[95%CI = 0.33–0.80]; >90,000, aOR=0.35, [95%CI = 0.21–0.57]), and not being in a relationship (aOR=0.63, [95%CI = 0.45–0.88]). While the finding for “other” ethnicity was statistically significant and associated with reduced odds of moderate/high CCA (aOR=0.57, [95%CI = 0.33–0.97]), this variable includes several ethnicities, including Arab, Black, and Filipino participants. Complete findings are available in the [Appendix \(Table A.4\)](#).

Interaction effects analyses ([Table 2](#)) revealed significant increases in CCA severity over time for individuals who were younger (16–24 years: aOR=1.44 [95 %CI = 1.17–1.77]; 25–44 years: aOR=1.26 [95 %CI = 1.07–1.47]), non-binary (aOR=1.94 [95 %CI = 1.39–2.81]), women (aOR=1.30 [95 %CI = 1.11–1.52]), sexually diverse (aOR=1.54 [95 %CI = 1.29–1.84]), not planning to have children (aOR=1.29, [95 %CI = 1.08–1.54]) or unsure about having children (aOR=1.72, [95 %CI = 1.37–2.17]), those with a disability (aOR=1.27 [95 %CI = 1.06–1.53]), lower annual income (<\$30,000; aOR=1.33, [95 %CI = 1.14–1.57]), those with at least a Bachelor's degree (aOR=1.23, [95 %CI = 1.05–1.43]), politically leaning more liberal (aOR per 1-unit increase=1.12, [95 %CI = 1.10–1.15]), and residing in high-density population areas (aOR=1.23, [95 %CI = 1.04–1.46]). Decreasing trends in CCA were observed among men (aOR=0.85, [95 %CI = 0.72–1.00]), individuals aged 65+ (aOR=0.63, [95 %CI = 0.49–0.80]), and those earning \$90,000 or more yearly (aOR=0.72, [95 %CI = 0.57–0.89]). Complete findings are available in the [Appendix \(Table A.5\)](#).

[Fig. 1](#) displays boxplots showing averages and trends of CCAS scores by survey wave for each sub-population. The boxplots for waves 2 (i.e., post-2021 heat dome) and 3 (i.e., post-2021 BCE floods) show higher median CCAS scores than wave 1, with wave 2 exhibiting the highest. For most groups, the smoothed loess regression lines in the boxplots indicate an upward trend in CCAS scores following both extreme weather events relative to wave 1 (i.e., pre-event data). Notably, only participants identifying as non-binary, of Indigenous ancestry, with slightly conservative political orientation, and those educated at a high school level or less have these regression lines continuing to rise after each event (indicating potential compounding CCA risks). In contrast, regression lines also suggest decreasing or stable CCA trends over time among older individuals (65+), South Asian and Chinese participants, participants earning over \$90,000 per year, and participants who are moderately or extremely conservative. A complete set of boxplots is available in the [Appendix \(Appendix, Figure 2\)](#).

4. Discussion

4.1. Scope and main results

This study explores the socio-demographic dimensions of climate change on mental health, focusing on factors associated with CCA among British Columbian adults following two successive extreme weather events in 2021. We found a complex set of socio-demographic factors associated with CCA. For instance, there was a higher risk of moderate to severe CCA among women, those unsure whether they want children, those with Bachelor's degrees or higher, and those with a more liberal political orientation, while lower CCA was associated with older age, higher income, and those not in a relationship. Compared to the pre-event period, CCA levels were generally higher after both events, peaking post-heatwave and subsequently declining. We only observed cumulative CCA increases due to successive event exposure in non-binary individuals, those with Indigenous ancestry, slightly conservative participants, and those educated at a high school level or lower. These findings signal that CCA is a broad eco-social phenomenon affecting diverse populations.

4.2. Interpretation and situation of results within the existing literature

This study's results align with previous research highlighting

Table 2

Estimates for the logistic regression models assessing interaction effects between socio-demographic factors and survey wave on climate change anxiety for British Columbians aged 16+ who were included in the analytic sample ($n = 1179$).

Interaction terms between socio-demographic covariates and survey wave ^a on CCA		aOR	95 % CI	p-value*
Wave X Age (years)	16–24	1.44	1.17 - 1.77	<0.001
	25–44	1.26	1.07 - 1.47	0.005
	45–64	0.94	0.79 - 1.11	0.470
	65+	0.63	0.49 - 0.80	<0.001
Wave X Gender	Man	0.85	0.72 - 1.00	0.047
	Non-binary	1.94	1.39 - 2.81	<0.001
	Woman	1.30	1.11 - 1.52	<0.001
Wave X Disability	No	1.04	0.90 - 1.20	0.593
	Yes	1.27	1.06 - 1.53	0.011
Wave X Sexual orientation	Heterosexual	0.97	0.83 - 1.12	0.666
	Sexually diverse	1.54	1.29 - 1.84	<0.001
Wave X Children	Yes	0.94	0.80 - 1.10	0.421
	No, and I am not sure	1.72	1.37 - 2.17	<0.001
	No, and I do not plan to	1.29	1.08 - 1.54	0.004
	No, but I plan to	1.21	0.98 - 1.49	0.074
Wave X Occupation	Trades, Natural Resources, Manufacturing	0.97	0.80 - 1.16	0.721
	Arts/Education/Government	1.41	1.17 - 1.70	<0.001
	Management/Business	1.07	0.87 - 1.33	0.507
	Not working	1.03	0.88 - 1.21	0.684
	Science/Health	1.18	0.97 - 1.45	0.104
Wave X Income	Less than \$30,000	1.33	1.14 - 1.57	<0.001
	\$30,000 to \$59,999	1.14	0.95 - 1.36	0.148
	\$60,000 to \$89,999	0.94	0.78 - 1.14	0.533
	\$90,000 or more	0.72	0.57 - 0.89	0.003
Wave X Education	High School or Less	1.01	0.84 - 1.21	0.932
	Some Post-Secondary Training	0.92	0.78 - 1.09	0.346
	Bachelor's degree or higher	1.23	1.05 - 1.43	0.009
Wave X Political orientation score		1.12	1.10 - 1.15	<0.001
Wave X Population Density/100/km²				

(continued on next page)

Table 2 (continued)

Interaction terms between socio-demographic covariates and survey wave ^a on CCA	aOR	95 % CI	p-value*
Low (<300/ km ²)	1.02	0.88 – 1.19	0.777
Medium (300 – 1499/ km ²)	1.08	0.89 – 1.29	0.440
High (>=1500/ km ²)	1.23	1.04 – 1.46	0.015

Abbreviations: aOR = adjusted odds ratio; CI = confidence interval; CCA = climate change anxiety; sexually diverse = sexual, bisexual, gay/lesbian, heterosexual, pansexual, queer, questioning.

^a a continuous numeric measure from 1 to 3 indicating the survey wave.

* numbers in bold are statistically significant to a p-value smaller than 0.05.

Note: Socio-demographic covariates included in the model were: survey wave, age, gender, ethnicity, disability status, sexual orientation, relationship status, child-raising intentions, occupation, income, education, population density of residential area, and political orientation.

increased mental health risks due to climate change among specific populations [7]. Education and income levels, proxies for

socioeconomic status, are regarded as key determinants of climate and mental health disparities [7]. Despite this, few studies have explored the relationships between education, income, life stage, and CCA. Contrary to a recent study [35], which found no differences in CCA by education or income, our study indicates that higher education increases the risk of moderate/high CCA, while higher income reduces this risk. Intriguingly, we also observed increasing CCA trends over time among people belonging to socially marginalized social categories such as lower education, non-binary, and Indigenous peoples. These discrepancies can be contextualized within the socioeconomic disparities in climate vulnerability literature [2]. While highly educated individuals may exhibit increased mental health resilience following natural disasters [36], they may also have heightened awareness of climate risks [37], contributing to higher overall CCA. In contrast, those with lower education may have limited exposure to climate change-related information and education [37], leading to lower initial levels of CCA. However, direct experiences with extreme weather events may catalyze their CCA trends over time [36]. From an income perspective, higher-income earners may have greater access to resources, including better housing infrastructure (e.g., air conditioning), increased social support, and a wider array of options for coping with extreme climate events [38,39]. Such resources may lower climate vulnerability risks and mitigate adverse climate impacts,

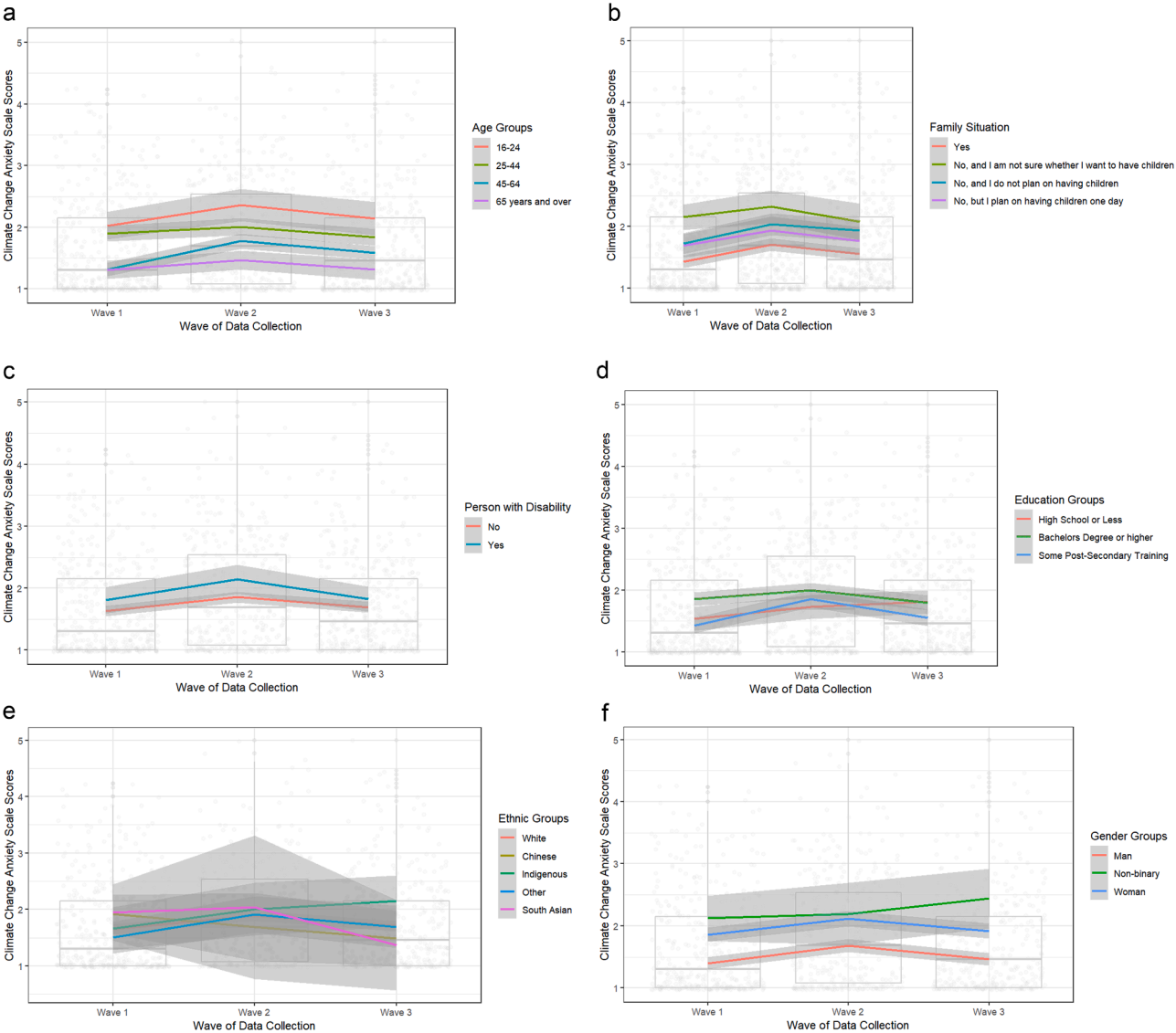


Fig. 1. Climate change anxiety trends over time across survey waves by socio-demographic characteristics.

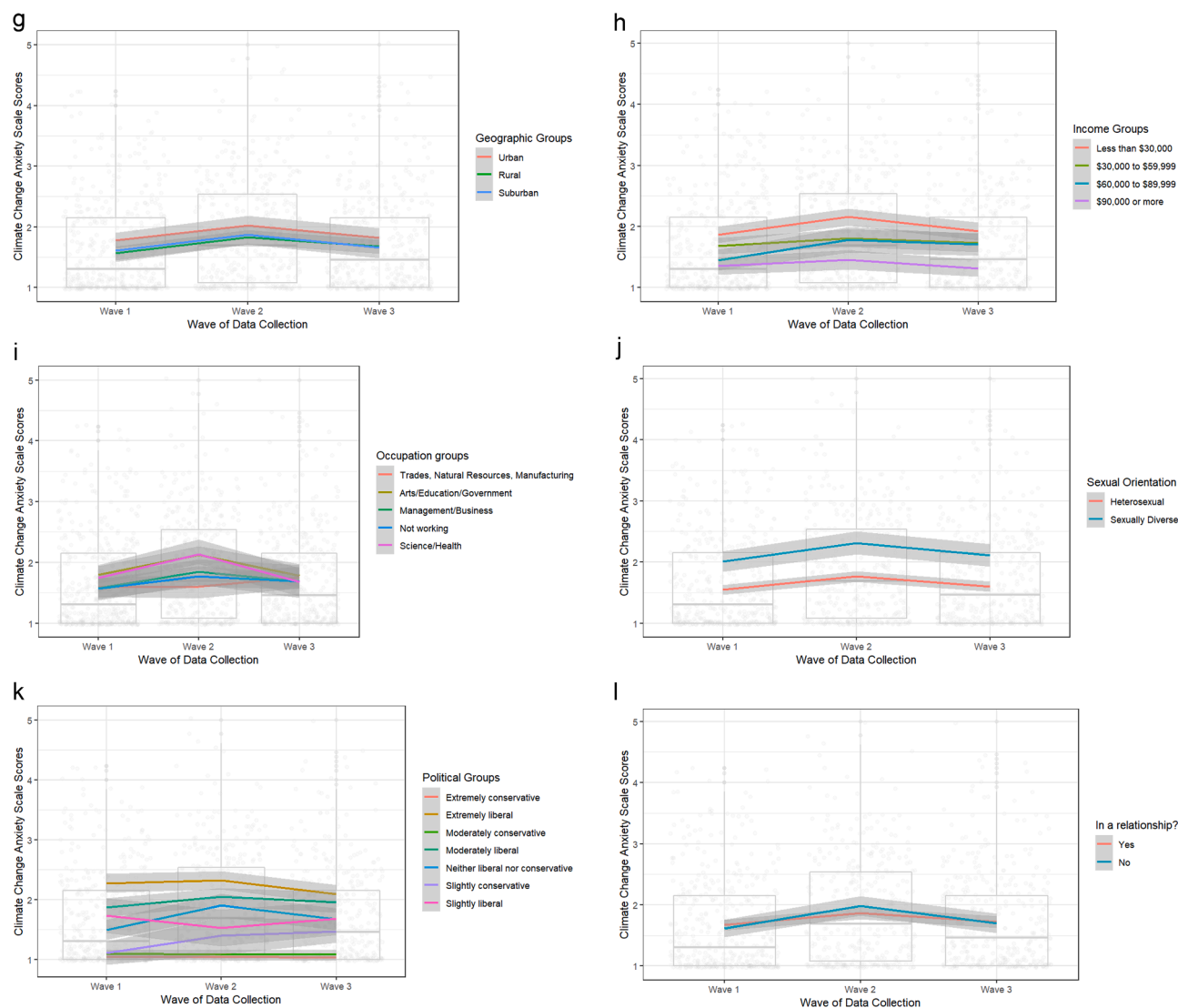


Fig. 1. (continued).

ultimately reducing anxiety levels [2]. Another potential mechanism is that high SES individuals have greater resilience and resources, leading to short-term effects but protecting them from long-term effects [2, 36–39]. In contrast, marginalized populations may not face immediate impacts but also have less capacity to bounce back [2,36,37].

Consistent with another study [35], we found that women were at a higher risk of experiencing increased levels of CCA than men. We also observed increasing CCA trends over time among non-binary individuals. Notably, non-binary individuals were one of only four groups in our study that exhibited a rise in CCA after each event; they were also the only one of these groups that were statistically significant in the interaction analysis, suggesting potential compounding CCA effects from successive exposures. These findings could indicate a potential lack of safety or comfort when accessing supports, with emergency shelters, evacuation programs, or similar programs not being trauma-informed, culturally sensitive, or queer positive [40,41]. These findings align with literature highlighting gender inequities as key contributors to disproportionate CCA and mental health burdens [42], and the overall exclusion of sexually and gender diverse persons from disaster management [43,44]. Women may generally experience heightened concerns about climate change and the future [45], and gender discrimination or gender-based violence are known risk factors for climate change vulnerability [42,46]. Other climate change mechanisms

disproportionately impact women, including prioritizing familial and communal care while putting their own needs aside [47,48]. Gender diverse persons may also have fewer social and economic resources to facilitate disaster recovery [43].

Our results showing higher levels of CCA among individuals unsure whether they want to have children contribute to the empirical evidence and public discourse on how climate concerns are impacting fertility decisions. One study found that 40% of 10,000 participants were reluctant to have children because of climate change [49].

Our study found a lower risk of CCA among adults aged 65+ than among those aged 16–24; this is interesting, as older adults are disproportionately affected by climate change and, thus, one might assume they would have higher levels of eco-anxiety [50]. Other studies corroborate this, with older adults having lower levels of climate anxiety compared to other age groups, despite being more vulnerable [50,51]. Older adults, particularly in the ‘Baby Boomer’ generation, have and continue to contribute disproportionately to carbon emissions, with some estimates suggesting the youth of today will emit 10 times less carbon than their grandparents in their lifetimes [52,53]. We also found decreasing CCA among older adults (65+) and increasing CCA among younger age groups (16–24) relative to baseline. This reflects generational differences in CCA, which may be attributed to financial insecurity or the long-term consequences of climate change that younger

generations face [49]; this may also heighten their sense of urgency and concern for future well-being [49]. Other factors that could impact CCA include political beliefs, morals, and personal choice, with older, more conservative individuals often having the lowest levels of CCA [54].

Individuals in our sample who were not in a relationship had a lower risk of severe CCA than those in relationships. Research suggests that romantic partners may influence each other's level of climate anxiety [55]. This finding may also be explained by the role of lifestyle-related factors in influencing climate resilience. Single individuals may have more decision-making autonomy or time for hobbies or exercise [56], which can improve overall mental health and alleviate climate change-related stress [57–59]. Additionally, single individuals may not be thinking about having children and thus may not have to worry about the effect that climate change could have on future generations. Research indicates that CCA can influence child-rearing intentions, as individuals are concerned about the world into which they are bringing their future children [60]. Also, parents who discuss climate change with their children and have climate-anxious children are more likely to have higher levels of CCA [61,62]. However, other studies show that when parents engage in meaningful dialogue and take climate action together, they can help lower concerns about climate change [63]. Being single has also been linked to higher mental health risks [64], in contrast to our findings. Thus, further research exploring the connections between relationship status and CCA is needed.

We also found that individuals with a more liberal political orientation were more likely to experience higher CCA, reflecting the alignment of current liberal political values with sustainability efforts [65]. Individuals with liberal ideologies are more likely to be concerned about climate change, potentially contributing to increased CCA [65]. In contrast, conservatism is linked to reduced climate concerns and skepticism about human contributions to global warming [65]. Interestingly, our study shows that slightly conservative individuals were the only politically related group to exhibit increasing CCA over time. This finding contrasts with previous research where conservative individuals showed increasing levels of climate denialism over time [66]. Our findings suggest that exposure to successive extreme weather events may have compounded CCA effects, revealing potential nuances in climate risk perceptions within the conservative spectrum. While these individuals may initially be uncertain about climate-related issues, their experiences with successive climate events may provide evidence of climate change impacts, prompting shifts in perspectives and enhancing their CCA over time, even if their overall CCA remains lower than more liberal participants.

We also found a statistically significant positive interaction between high-density geographic areas, wave number, and CCA, contrasting with research emphasizing greater climate vulnerability among rural communities [2]. It is plausible that urban residents may have increased exposure to information about climate impacts, raising their awareness and concerns and leading to elevated CCA [37]. Additionally, urban dwellers may perceive themselves at greater risk from climate impacts, with less resilience, given the potential for more severe consequences in densely populated areas with concentrated infrastructures [67]. People who are urban poor also deal with intersecting concerns, including poor housing and working conditions, and worse access to health care [68]. However, this situation is nuanced. Rural communities are more sensitive to climate effects, such as vegetation changes and increased extreme weather events, with limited support infrastructure and transportation, and a greater reliance on natural resources [69]. While urban dwellers may not see direct impacts on vegetation, they are likely to see the effects of climate change through urban heat islands [70].

The results indicate a significant increase in the risk of more severe CCA during the 2021 North American heatwave compared to the pre-heatwave period. While CCA levels were higher after each of the two weather events than pre-event levels, we observed an increasing trend following the heatwave, which decreased rather than continuing to rise over the following five months. These findings can be explained by

differences in the scale and severity of each event's impact. The heatwave event was unprecedented, affected a larger area, and impacted more people; it was also associated with 750 heat-related deaths in BC [14], a substantially higher mortality rate than the November floods. Furthermore, a possible ceiling effect might have limited further CCA increases in response to the floods after the heatwave.

4.3. Strengths and limitations

Our study has several strengths, including its novel focus on socio-demographic factors associated with moderate/high CCA in BC and its contribution to a largely unexplored Canadian climate vulnerability research field. Our findings advance the understanding of differences within and between population groups whose mental health is affected by climate change. Future research could look at social processes of marginalization beyond socio-demographic variables (e.g., sexism, racism, LGBTQ stigma) to inform intersectional approaches that examine multiple interlocking systems of oppression [71,72]. Such knowledge could enhance targeted adaptation strategies by delineating high-impact risk and opportunity focus areas. The study's methodological strengths include (a) using quasi-experimental data collection before and after two successive extreme weather events, which mitigates history-related threats to internal validity; (b) using the CCAS scale as a validated instrument with strong psychometric properties for collecting outcome data, reducing risks of measurement error bias; and (c) the variety and richness of data collected, spanning across 12 different socio-demographic covariates of interest, which permitted a comprehensive examination the variety of potential socio-demographic factors influencing CCA in the BC context.

A notable limitation of the study is its repeated cross-sectional design, which prevents longitudinal assessment of within-person changes in CCAS scores. This is particularly relevant in the context of mental health impacts of extreme weather events, as these involve complex interactions between vulnerability and exposure factors [4]. Longitudinal explorations would provide valuable insights into the evolution of CCA over time and the impact of exposure to different climate stressors. Due to this limitation, we cannot be certain whether the findings are attributed to the weather events, third-variable effects, time, or sample variability at the time of the assessment, given that we did not survey the same people at each wave. Future studies could implement a longitudinal design to assess how CCA develops within individuals over time and across climate events. Another limitation is reliance on self-reported data and proxy measures for factors of interest, which may be subject to information bias due to recall, response, or social desirability bias (e.g. measuring exposure to extreme weather events or climate stressors before the study).

This study used pre-existing data, so we could not conduct a power analysis before the study. Post-hoc power analyses have been largely discouraged by previous literature [73–75]. Regardless, we have conducted a post-hoc power analysis (Appendix S1). Thus, interpretation should be approached with caution, particularly when findings are of borderline significance. While our sample was representative of BC in terms of proportions of participants' gender, age, income, and population density [76], our analytical sample predominantly included White, highly educated adults with more liberal political orientations. This indicates a lack of equitable representation of participants across communities most affected by climate change, limiting external validity. Due to uneven sample demographics, the generalizability of the findings is also limited. As certain demographics may respond in a particular way, and because populations are unevenly represented within our study, there is a chance that results can be over- or under-estimated [77]. The absence of Indigenous peoples in the sample also limits broad interpretation, while applying a distinctions-based approach in future work could allow further exploration into the unique experiences of different Indigenous groups in Canada, including the First Nations, Inuit, and Metis [78].

Despite these limitations, the robustness of our analytic approach aims to offset potential concerns arising from the lack of a power analysis and the limited number of socio-demographic variables. We therefore recommend that future studies in this field, especially those exploring subtle effects or interactions, prioritize securing larger sample sizes or consider collaborative multi-site efforts to ensure adequate power. This exploratory analysis can serve as a rationale or a feasibility test for costly, truly generalizable and representative approaches.

Other aspects of the study design could be considered limitations. For example, while this study examines the nuances of CCA vulnerabilities, it does not analyze intersectional realities (e.g. social processes of sexism, racism). This study also loses some nuance as certain demographic categories are grouped, particularly ethnicity (classified as 'Other') and sexuality (classified as 'Sexually Diverse'). Additionally, the interpretation of the income finding in this study could overlap, as approximately 34 % of those considered low-income were not working. Future studies can further examine multiple social processes of marginalization in CCA, enabling an in-depth exploration.

This study discusses a topic in the same field as a previously published study by this research lab [14]. However, the focus of this study differs from the previous study in two key aspects: a) it assesses the cumulative effects of several extreme climate events on CCA, whereas the previous study examined the effect of a single climate event on CCA; and b) it investigates some socio-demographic differences in climate change anxiety, whereas the previous study did not.

Also, as this data was collected during the COVID-19 pandemic, it is possible that anxiety levels were generally higher. Research is mixed, with some studies showing a positive correlation with COVID-19-related anxiety and CCA, while others show decreased levels of CCA in older adults [79,80]. Other studies show youth being more concerned about the climate crisis than COVID-19 [81]. Future studies could use newer datasets that do not need to consider (or consider to a lesser extent) COVID-19 as a potential confounding factor.

4.4. Implications and recommendations

Our study underscores the importance of incorporating socio-demographic perspectives into climate change and mental health research. In addition to addressing gaps in the Canadian CCA-related literature, our findings also reveal disparities in mental health across communities at risk of climate impacts, contributing to broader climate vulnerability research. We highlight the need to consider interaction effects between socio-demographic factors and exposure to climate-related events to better understand the unique experiences of diverse groups over time. These insights can guide targeted interventions, inform equitable climate policies, and facilitate more holistic public health interventions that address specific needs and vulnerabilities, ultimately bridging gaps between climate adaptation and mitigation efforts and public health. Notably, our results support expert recommendations to consider socio-demographic factors within a broader climate vulnerability context to enhance adaptation-related decision-making [2].

Future research should examine the interplay between socio-demographic and geographic dimensions of CCA to develop equity-focused data collection strategies for climate vulnerability research. Equity-driven data facilitates tailored health risk assessment and management actions while characterizing power structures to help enable intersectional assessments of climate change impacts [42]. Future research should also explore innovative data-collection methods that involve climate-affected communities and promote power-sharing and community research participation across different groups. As climate vulnerability research requires multidisciplinary approaches, this can also foster capacity building and intersectoral collaborations to support adaptation efforts in Canada. Future research can also examine the intersectional relationships between social categories and social processes of marginalization to better understand the nuances of CCA

vulnerabilities, assess climate resilience, coping, and assets across multiple marginalized communities, and highlight resistance and strengths [72].

5. Conclusion

This study examined the socio-demographic factors associated with CCA and its trends among British Columbians after two extreme weather events in 2021. The findings revealed higher CCA among women and non-binary individuals, those unsure whether they wanted children, those who were highly educated, and liberal-leaning individuals, while older adults, high-income earners, and those not in a relationship experienced lower levels of CCA. The research highlights the importance of equity-focused approaches in climate vulnerability research, policy development, and mental health interventions to support at-risk communities. Further investigation is essential to elucidate causal relationships and the underlying mechanisms driving these associations.

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Ethics approval

The study received ethical approval from the Research Ethics Board (REB) at Simon Fraser University (SFU) (REB#: 30000309).

Consent: Participants provided informed consent prior to study participation.

Availability of data and material: Data is available upon request.

CRediT authorship contribution statement

Andreea Bratu: Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Conceptualization. **Aayush Sharma:** Writing – review & editing. **Kalysha Closson:** Writing – review & editing. **Maya Gislason:** Writing – review & editing. **Robert S. Hogg:** Writing – review & editing, Funding acquisition. **Carmen H. Logie:** Writing – review & editing. **Carly Marshall:** Writing – review & editing. **Gina Martin:** Writing – review & editing. **Tim K. Takaro:** Writing – review & editing. **Meghan Wise:** Writing – review & editing. **Kiffer G. Card:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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